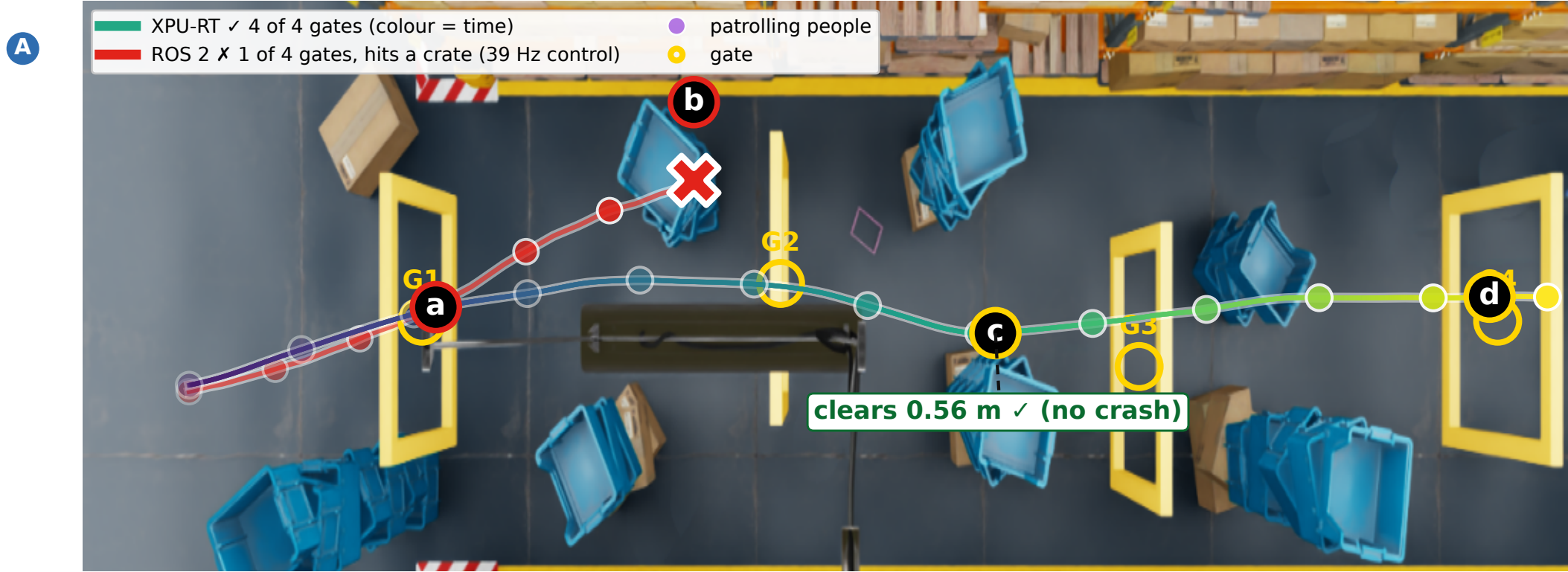
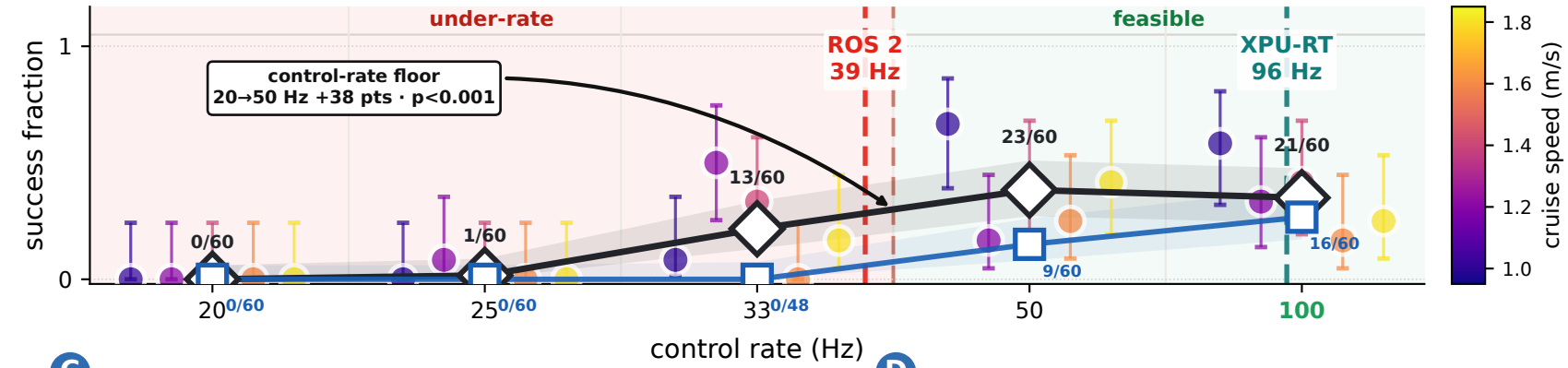


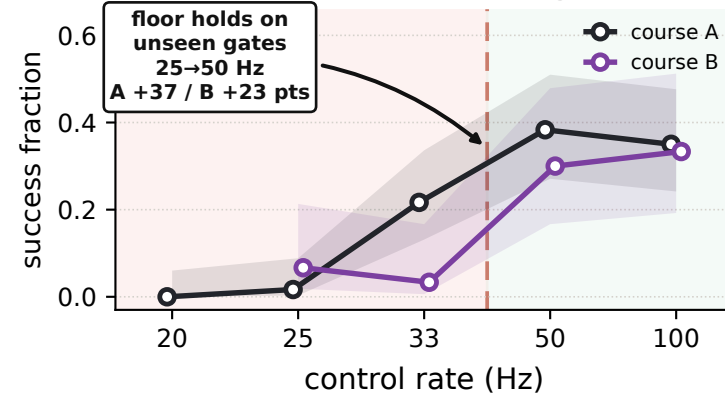
Warehouse gate-course showdown — same scene and drone, gain, at 1.4 m/s
XPU-RT · CP-SAT (96 Hz control) clears all 4 gates
ROS 2 vanilla (39 Hz control) clears G1 → hits a crate
one seed of this scene (seed 1003); over its 12 seeds XPU-RT completes 4 and ROS 2 none



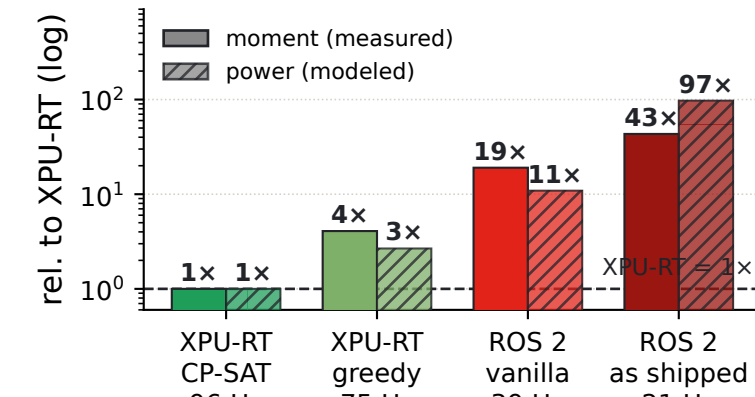
B Control-rate envelope (rate injected, 300 flights): success against control rate — ROS 2's 39 Hz and XPU-RT's 96 Hz marked



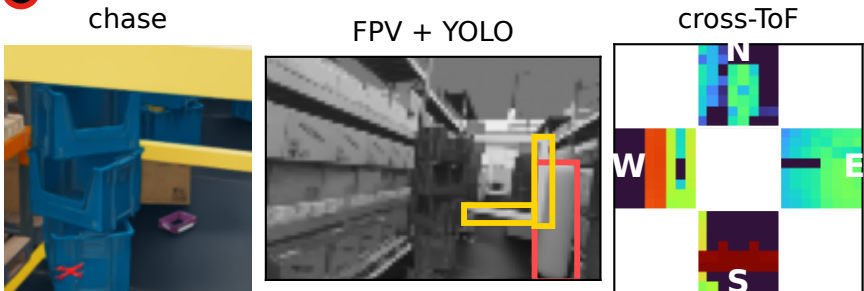
C Generalization: unseen gates (course B)



D Mechanism: the baseline thrashes (n=6 flights/arm)



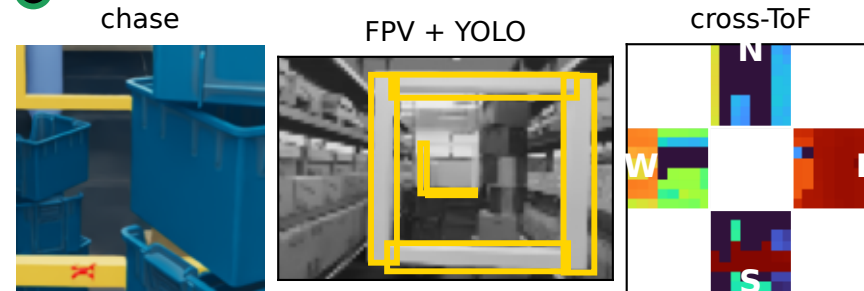
a ROS 2 39 Hz · clears G1 · t=2.9s



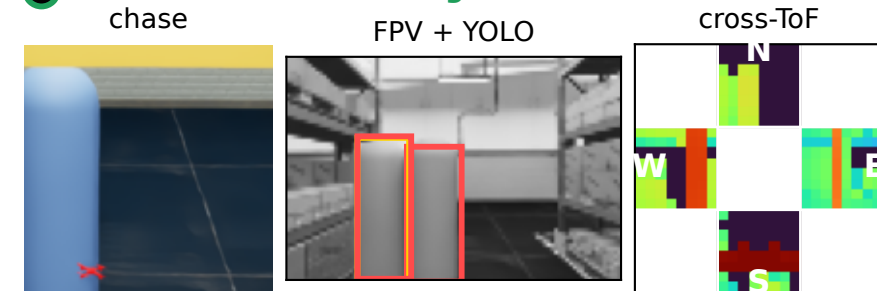
b ROS 2 39 Hz · hits crate after G1 · t=5.0s



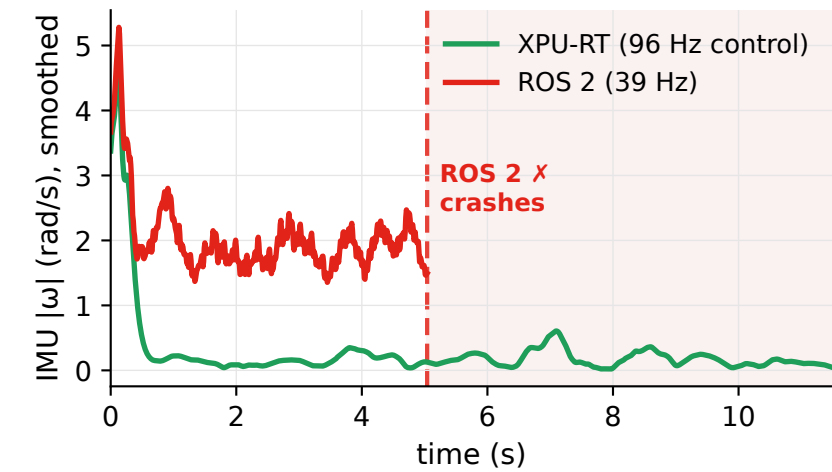
c XPU-RT 96 Hz · clears crate 0.56 m · t=7.2s



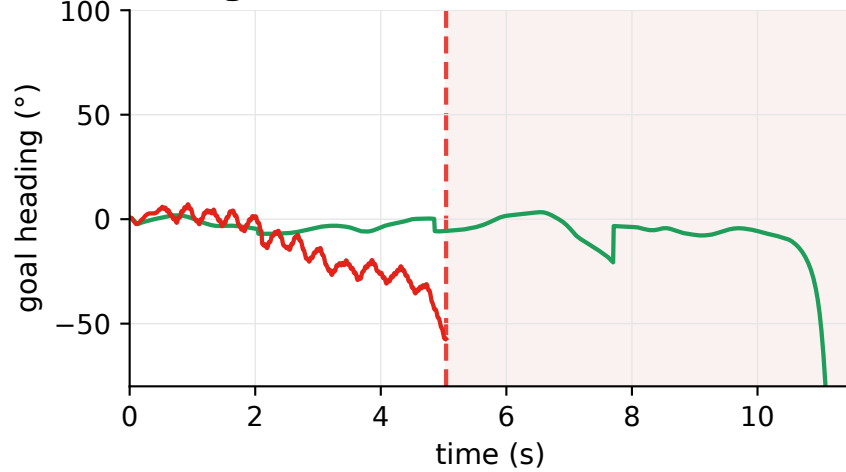
d XPU-RT 96 Hz · reaches gate · t=11.0s



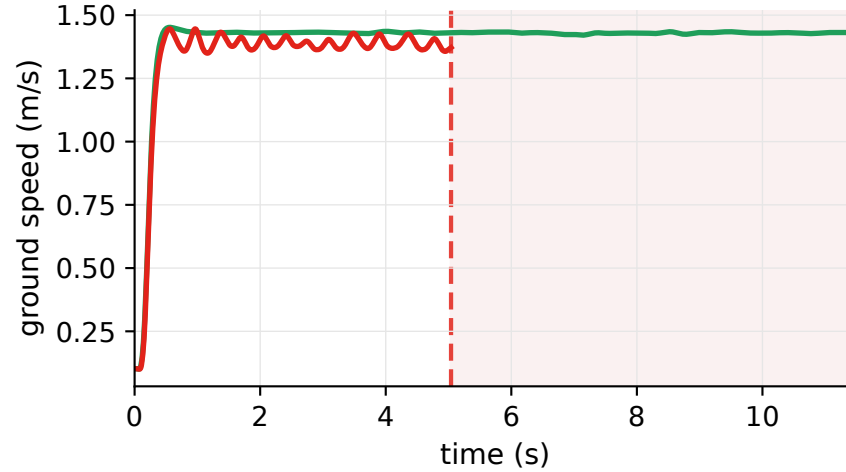
E Inside the pair — body rate: the baseline thrashes at 39 Hz



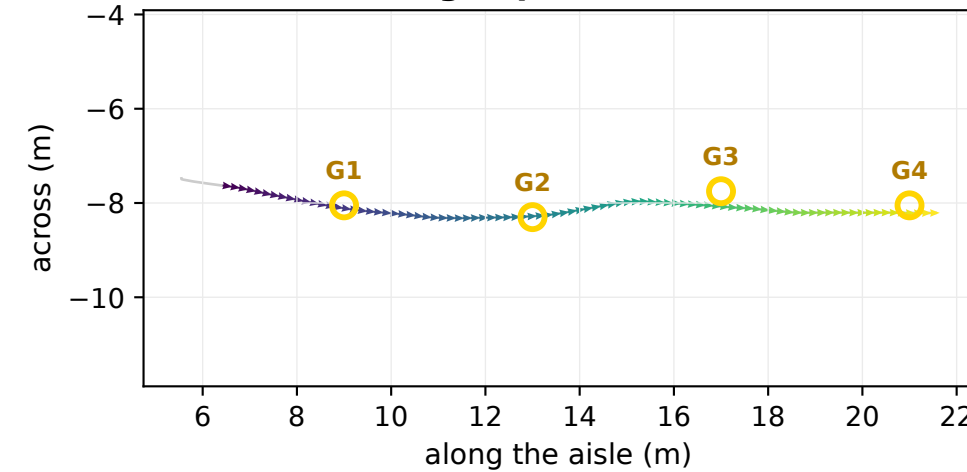
F Nav goal heading: the two commands diverge before the baseline's collision



G Forward speed: the same cruise until the baseline's collision



H XPU-RT's velocity through the gates (arrow = heading · speed, colour = time)



Onboard K1 schedule, measured at the 45 Hz camera — XPU-RT · CP-SAT places the kernels over all 8 harts: camera→control 57 ms, control every 10.0 ms, every frame on time · YOLO on 8 harts, control on 8 harts, nav on 7 harts
ROS 2 (4 processes · single executor · chained control · unpinned): camera→control 242 ms, control every 26.0 ms, 760/766 frames late · YOLO on 4 harts, nav on 1 hart, control on 1 hart

ROS 2 · all 8 cores (5 processes · single executor · chained control · unpinned): camera→control 37 ms, control every 22.2 ms, 0/448 frames late · YOLO on 5 harts, nav on 2 harts, control on 1 hart
bars: a hart's consecutive dispatches of one frame joined, idle gaps kept — the same unit of work as a ROS 2 callback (545 XPU-RT dispatches in this window, 409 bars)

